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Techno-economic feasibility analysis of a hybrid wind-PV-diesel energy system for Kiltu village, Oromia, Ethiopia using HOMER pro

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Abstract

Indoor air pollution from biomass fuel use poses a significant health threat, especially for women and children in rural communities, due to its widespread use for cooking and heating. This study evaluates the feasibility and cost-effectiveness of a hybrid energy system (solar, wind, and diesel) for Kiltu Village, Oromia, using HOMER Pro software to optimize technical, economic, and environmental outcomes. Key factors include an average daily solar irradiation of 6.06 kWh m^{-2}, wind speed of 4.61 m s^{-1}, and a total daily energy demand of 931.44 kWh, with a peak load of 115.19 kW. Simulation results show that a PV-diesel-battery-converter system is the most efficient, with a levelized cost of electricity (LCOE) of \$0.326/kWh and a net present cost (NPC) of \$1.42 million, reducing CO_2 emissions by 49.16% compared to diesel-only systems. Although the fully renewable solar-wind-battery system has higher costs (NPC: \$2.10 million), the hybrid system significantly cuts diesel consumption and meets essential seasonal agricultural demands, such as winter irrigation. This hybrid solution offers a sustainable way to reduce indoor air pollution, strengthen energy resilience, and promote socio-economic development in rural areas. The findings highlight hybrid energy systems as an effective approach for improving public health and advancing sustainable electrification in energy-poor regions.

Abbreviations

Abbreviations	Definitions
COE	Cost of Energy
DG	Diesel Generator
HES	Hybrid Energy System
HOMER	Hybrid Optimization Model for Electric Renewables
IEA	International Energy Agency
kW	kilo Watt
kWh	killo Watt hour
LCOE	Levelized cost of energy
NPC	Net Present Cost
NPV	Net present value
PV	Photovoltaic

RF	Renewable Fraction
SSA	Sub-Saharan Africa
WHO	World Health Organization

Key findings

A tailored hybrid energy system was developed for Kiltu Village, ensuring a stable and continuous power supply throughout the year.

The system integrates solar, wind, and diesel energy sources with battery storage to provide consistent off-grid electricity.

A techno-economic analysis conducted using HOMER Pro identified the most optimal configuration, achieving a 49.16% reduction in CO_2 emissions.

The system replaces biomass usage with clean energy alternatives, improving indoor air quality and overall health.

A scalable approach is proposed for optimizing energy solutions, considering local energy demands and resource availability.

1. Introduction

In the 21st century, global energy demand has surged owing to economic growth, population expansion, and industrial development (Rehman *et al* 2016). Access to energy is fundamental for socio-economic progress, directly influencing quality of life and community well-being (Amole *et al* 2023). However, approximately 1.3 billion people, primarily in rural areas, lack reliable electricity due to low population density, insufficient infrastructure, and economic constraints (Das *et al* 2021). Sub-Saharan Africa, where 60% of the population resides in rural areas, remains the epicenter of energy poverty. Here, rural electrification is both urgent and challenging, as only 5% of the population has access to clean energy (Akinyele *et al* 2018, Alanne 2021, Babayomi *et al* 2023, Avordeh *et al* 2024).

To address this crisis, the International Energy Agency (IEA) has promoted small-scale off-grid systems as cost-effective solutions for rural electrification, laying the groundwork for future energy networks (Patel and Singal 2018, Mulenga *et al* 2023). Advances in renewable energy technologies, particularly solar and wind, have further enhanced the feasibility of these systems. For example, solar energy costs declined dramatically from \$0.38/kWh in 2010 to \$0.06/kWh, while wind energy costs fell from \$0.09/kWh to \$0.04/kWh kWh (Karamov *et al* 2022). These technological advancements are crucial for improving health and quality of life in low-income countries, where dependence on biomass fuels for cooking exposes rural populations to harmful indoor air pollution (Adeoye *et al* 2024).

Household air pollution is particularly prevalent in sub-Saharan Africa, where unclean fuels are linked to higher rates of respiratory issues among children, including coughing and rapid breathing (Amadu *et al* 2023, Panchal *et al* 2024). A 10 $\mu g\ m^{-3}$ increase in PM2.5 concentration has been associated with a 6% rise in acute respiratory infections (ARI) in children under five, with rural populations and younger boys being the most affected (Odo *et al* 2022). Therefore, transitioning to clean energy significantly improves health outcomes, particularly for women and children, while boosting life expectancy and education(Nnuka Tsekane *et al* 2024). Off-grid solar-powered induction cookers, for example, offer eco-friendly alternatives, achieving 11.8% energy efficiency and operational costs as low as \$3.88/month (Altouni *et al* 2022). Standalone PV mini-grids with battery storage now offer affordable and sustainable solutions for remote communities, promoting energy access, sustainability, and economic growth (Come Zebra *et al* 2021, Gebreslassie *et al* 2022). Hybrid renewable energy systems (HES) have proven to be a successful solution for meeting energy needs while promoting sustainability globally. For instance, in Bangladesh, a hybrid system satisfies a daily demand of 1,000 kWh, generates 4,604.59 MWh annually, and achieves a LCOE of 0.03 USD/kWh. This system provides a 6.6-year payback period, reduces CO_2 emissions by 5,767 tons (89%), and offers substantial cost savings, enhanced climate resilience, and effective grid integration (Ashraful Islam *et al* 2024). Similarly, a study on a standalone solar/wind hybrid system in Giri, Nigeria, conducted using HOMER, identified an optimal configuration with a NPC of 1.01 million USD, a COE of \$0.110/kWh, and a renewable energy fraction of 98.3%. The system also achieves an annual GHG reduction of 2,889 kg, demonstrating its cost-effectiveness and environmental benefits (Salisu *et al* 2019). In Jordan, a hybrid PV/wind/grid system provides electricity at a cost of \$0.024/kWh, with 71.8% of the energy coming from wind and 28.2% from the grid. This system reduces CO_2 emissions by 220 tons per year (53%) and contributes significantly to improving energy security and sustainability (Al Afif *et al* 2023). In Nigeria, HOMER Pro simulations show that a PV-Diesel system can save \$30,583 annually, with a NPV of

\$390,949 and a payback period of just 1.3 years, while reducing CO_2 emissions by 75 tons. In comparison, a Wind-Diesel system saves \$15,174 per year, with an NPV of \$193,980 (Yakub *et al* 2022). Moreover, a hybrid energy system in Ghana, combining Solar PV, Diesel Generators, and Battery Storage, was optimized in HOMER to demonstrate an LCOE of \$0.0263/kWh, a renewable fraction of 91.5%, and an NPC of \$244,937. This configuration highlights its viability over traditional grid electricity for remote communities, reinforcing the potential of hybrid systems in rural electrification (Yakubu *et al* 2025).

Ethiopia faces a significant energy challenge, with 85% of its population living in rural areas, yet only 1% having access to electricity (Solomon *et al* 2022). This energy poverty forces heavy reliance on traditional biomass fuels, which account for 90% of household energy consumption (Liyew *et al* 2024). The use of biomass exacerbates health risks, environmental degradation, and CO_2 emissions (Yalew 2022). Household air pollution remains severe, disproportionately affecting vulnerable groups such as women, children, and pregnant women (Okello *et al* 2018, Zhou *et al* 2024). Indoor air pollution, which exceeds WHO limits, contributes to respiratory illnesses, particularly among young children and pregnant women (Asefa and Mergia 2022). Both indoor (cooking fuel) and ambient pollution (e.g., NOx, NO_2) are prevalent in Ethiopia (Flanagan *et al* 2022).

Despite these challenges, Ethiopia, located in the tropical zone, receives abundant average solar radiation ranging from 5 to 8 kWh/m^2/day (Woldegiyorgis 2019, Woldegiyorgis *et al* 2023a, 2023b). Studies in Northeastern Ethiopia (Kemissie, Haik, and Woldiya) have shown PV cell energy outputs between 0.1101 and 0.1844 kWh, confirming the region's suitability for solar (Woldegiyorgis *et al* 2024a). Similarly, PV systems near Fiche town demonstrate significant potential for electricity generation, offering solutions to indoor pollution in nearby rural areas (Woldegiyorgis *et al* 2023a, 2023b). Wind energy also holds great potential for off-grid electricity in remote areas; however, technical and economic challenges hinder its full utilization (Benti *et al* 2021, Woldegiyorgis *et al* 2024b). Meanwhile, HOMER-based studies have demonstrated the feasibility, cost-effectiveness, and emissions reduction potential of hybrid PV/Wind/Diesel/Battery systems (Das *et al* 2021, Asghar *et al* 2023). Successful implementations in Wollega, Maji Town Zone, and Oromia highlight their viability across Ethiopia's diverse climates (Woldegiyorgis 2019, Benti *et al* 2023, Tuleman 2023). Such systems not only reduce CO_2 emissions but also address critical socio-economic challenges. For instance, PV systems improve household incomes and expand educational opportunities for students (Wassie and Adaramola 2021).

According to previous studies, rural electrification in Ethiopia faces technical, economic, and infrastructural challenges, necessitating scalable off-grid solutions. The study site lacks access to the national grid, and unreliable electricity disrupts critical activities such as winter wheat and onion farming due to inadequate irrigation, limits students' ability to study, and renders essential office equipment nonfunctional highlighting the far-reaching consequences of energy insecurity. Meanwhile, dependence on biomass fuels exacerbates indoor air pollution, posing significant health risks, particularly for women and children. While clean energy could mitigate these risks, its health benefits remain underexplored. Hybrid energy systems integrating solar PV, wind, diesel, and battery storage show promise but remain unoptimized for rural areas like Kiltu village. Existing studies largely overlook the role of renewable energy in reducing indoor pollution and lack comprehensive data on the performance of hybrid systems in rural communities. This study evaluates the feasibility and cost-effectiveness of hybrid systems incorporating solar PV, diesel, and wind turbines for Kiltu village using HOMER Pro.

2. Materials and methods

2.1. Study area and its description

The study was conducted in Kiltu village, located near Dera town in Oromia, Ethiopia, approximately 5 kilometers from Dera and in proximity to Dilfekar Park within the Arsi Mountains National Park. This remote community faces severe energy poverty, which significantly hampers development across multiple sectors.

The heavy reliance on biomass fuels such as wood, charcoal, and dung for cooking and heating greatly intensifies indoor air pollution, disproportionately affecting women and children, while also contributing to deforestation and environmental degradation. These issues are part of a broader ecological crisis, linking local energy challenges to global climate change. Furthermore, the village is situated in one of Oromia's drought-prone regions, which increases its vulnerabilities and highlights the urgent need for reliable energy access. Despite these challenges, the region offers considerable potential for renewable energy development, thanks to its abundant solar and wind resources. Transitioning to clean energy is not only a feasible solution but a vital step toward long-term sustainability. An integrated approach that combines solar and wind power with energy storage can foster significant improvements, benefiting public health, economic stability, and environmental conservation.

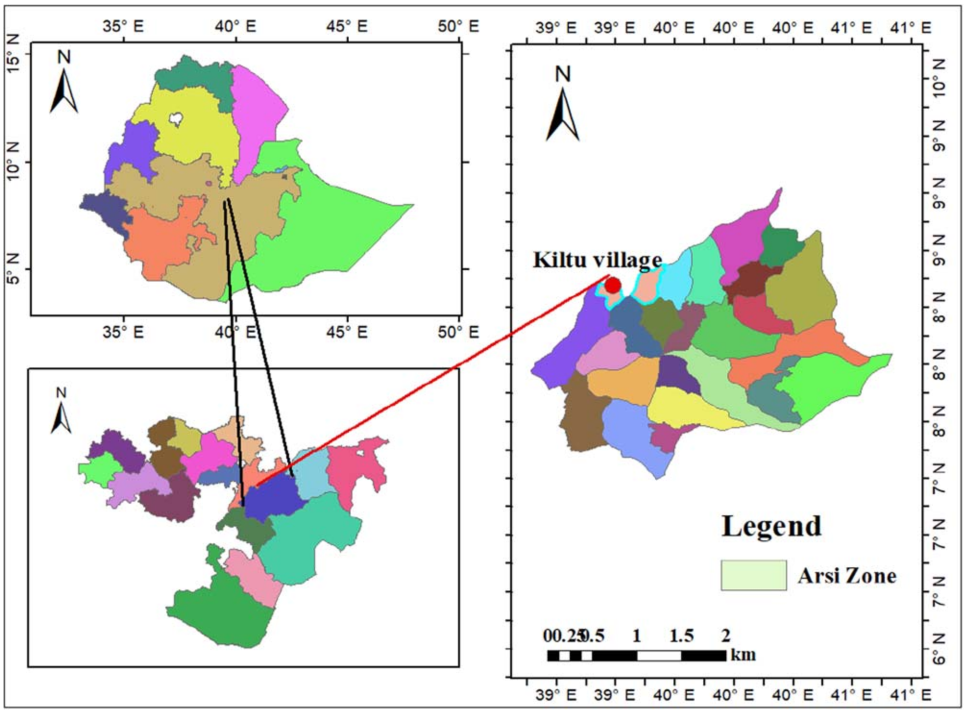

Figure 1. Map of Kiltu village.

2.2. Data collection and system design

The study began with the collection of site-specific data to lay the groundwork for designing the energy system. Data on solar irradiance and wind speed, sourced from NASA, were analyzed to assess the renewable energy potential of the site, while diesel supply information was integrated to account for the availability of existing or backup energy sources. At the same time, energy consumption profiles were created for 400 households, along with key community institutions such as a school, church, and administrative office. These profiles captured both daily and seasonal consumption patterns, ensuring that the proposed hybrid energy system aligned with the community's actual energy demands. By combining resource availability with accurate demand data, the procedure ensured a stable approach that met both technical feasibility and the community's energy needs.

Additionally, HOMER Pro evaluated environmental impacts, including CO_2, CO, SO_2, NOx, particulate matter, and unburned hydrocarbons, to balance cost with sustainability. Sensitivity analysis was used to assess different factors, such as fuel prices and resource availability, to examine system performance under various scenarios and ensure adaptability to future uncertainties. A comparative analysis with standalone diesel generators was conducted to explore the trade-offs between cost, renewable energy integration, and emissions. The optimization process aimed to minimize NPC/LCOE and emissions while maximizing RF. This methodology provided a balanced, data-driven approach to delivering reliable, affordable, and eco-friendly energy access.

2.3. Mathematical modeling of standalone renewable energy systems

2.3.1. Modeling of photovoltaic systems

Photovoltaic systems convert sunlight into DC power by capturing energy from photoelectrons. Power output depends on solar irradiation and cell temperature. Crystalline silicon panels are the most common technology, with output calculated as shown in equation (1) (Nsafon *et al* 2020, Sadeghi *et al* 2024).

$$P_{pv}(t) = (P_R)(D_{PV})\left[\frac{GHI(t)}{GHI_{ref}}\right][1 + \alpha_p(T_{cell}(t) - T_{ref})] \tag{1}$$

Where P_R is the derating factor, D_{PV} is the percentage of panel performance, and GHI (t) is the solar irradiance on the panels (kW/m^2). The reference irradiation GHI_{ref} is 1000 W m^{-2}, the temperature coefficient is α_p, and T_{cell} represents the temperature of the solar cell.

2.3.2. Modeling of diesel generator systems

Fossil fuels, mainly diesel, power internal combustion engine generators used for energy due to availability and low cost. Optimal sizing minimizes generator runtime, used as backup when other sources can't meet the load.

Fuel consumption is calculated using the following equation (Lata-García *et al* 2024, Mashrur *et al* 2024, Mohamed *et al* 2024)

$$C_{DG} = f_1(R_{DG}) + (P_{DG})f_2 \tag{2}$$

Where C_{DG} is the fuel consumption in liters, f_1 is the fuel curve coefficient in liters per hour, R_{DG} is the diesel generator's nominal power, and P_{DG} is the generated power in kW.

2.3.3. Modeling of wind turbine systems

The electrical output of a wind turbine depends on factors like turbines, wind speed distribution, conversion efficiency, and maintenance (Olatomiwa *et al* 2015, Emmanuel *et al* 2020, Ngila and Farzaneh 2023, Prum *et al* 2024). Other factors affecting electrical power output include tower height and the power output curve (Emezirinwune *et al* 2024). A turbine harnessing wind's mechanical energy to generate electricity under standard pressure and temperature conditions can be described by equation (3) (Das *et al* 2021, Sadeghibakhtiar *et al* 2024):

$$P_{WT}(t) = \begin{cases} 0, \ v(t) \leqslant v_{cut-in} \text{ or } v(t) \geqslant v_{cut-out} \\ P_r\left[\frac{v^3(t) - v^3{}_{cut-in}}{v_r{}^3 - v^3{}_{cut-in}}\right], \quad v_{cut-in} < v(t) < v_r \\ P_r, \quad v_r \leqslant v(t) \leqslant v_{cut-in} \end{cases} \tag{3}$$

where P_r, v_r, $v_{cut\text{-}in}$, and $v_{cut\text{-}out}$ denote the rated power, rated velocity, cut-in velocity, and cut-out velocity of the turbine. Wind velocity varies with altitude and the height of measurement. P_{WT} represents turbine power based on wind speed, determined from hub-height wind speed (Wang *et al* 2024).

$$v(t) = v_{ref}\left[\frac{H_{WT}}{H_{ref}}\right]^{\alpha} \tag{4}$$

The reference height H_{ref} and turbine hub height H_{WT} are crucial for wind energy calculations. The wind shear coefficient, or Hellmann exponent, is represented by the dimensionless parameter α, ranging from 0.1 to 0.4 (Fosso Tajouo *et al* 2023).

2.3.4. Modeling of battery storage systems

Excess energy from renewable sources is stored in batteries for use during shortages (Oladigbolu *et al* 2021). The storage capacity is determined using the following equation (Lata-García *et al* 2024).

$$E_{Batt}(t) = E_{Batt}(t-1) + E_{Excess}(t)(\eta_{Batt})(\eta_{inv}) \tag{5}$$

$$E_{Excess}(t) = [E_{AC}(t) + E_{DC}(t)\eta_{inv}] - E_{Demand}(t) \tag{6}$$

Where E_{Batt} is the energy stored in the batteries at a given state, $E_{Batt}(t-1)$ is the energy stored in the previous state, η_{Batt} is the battery efficiency, η_{inv} is the inverter efficiency, E_{AC} is the inverter output energy, and E_{DC} is the generated energy.

2.3.5. Power system inverters

The inverter connects all DC generation sources to the AC load, converting energy flow. With over 95% efficiency, it is sized according to the system's peak power, and its power is determined using equation (7) (Agajie *et al* 2023, Lata-García *et al* 2024).

$$P_{inv} = \frac{P_{peak}}{\eta_{inv}} \tag{7}$$

Where P_{inv} represents the inverter's power, P_{peak} is the total peak load power of the system, and η_{inv} is the component's efficiency.

2.3.6. Financial evaluation

In the present study, the optimal result is determined based on the lowest COE and the NPC of the system configuration. The HOMER pro software tool determines the COE using equation (2) (Konneh *et al* 2021, Mulenga *et al* 2023).

$$COE = \frac{C_{ACC} + C_{ARC} + C_{AOM}}{E_{AES}} \tag{8}$$

Where C_{ACC}, C_{ARC}, and C_{AOM} are the annualized capital cost, annualized replacement cost, and the annualized operation and maintenance cost, respectively, and E_{AES} is the annualized electricity served.

Table 1. Load profile of Kiltu village.

Service type	Load type	Rated power (w)	Quantity	Running time (h)	Energy (Wh/day)	(kWh/day) of village
Households (400)	Light	11	6	5	1320000	132.00
	Radio	100	1	11	440000	440.00
	TV	70	1	6	168000	168.00
	Charging mobile	5	20	3	120000	120.00
	Iron-box	1000	20	2	40000	40.00
	Other	300	1	5	1500	1.500
Subtotal						901.5
Church	CFL lamp	15	8	8	960	0.96
	Tape recorder	50	2	8	800	0.8
	Microphone	75	1	10	750	0.75
	Computer	100	2	6	1200	1.2
	Others	100	1	10	1000	1.0
Subtotal						4.71
Administration office	Desktop	100	2	8	1600	1.6
	Printer	40	1	6	240	0.24
Subtotal						1.84
School	CFL lamp	15	15	10	2250	2.25
	Desktop	100	20	8	16000	16.0
	Microphone	75	1	4	300	0.3
	TV	250	1	4	1000	1.00
	Tape recorder	50	6	8	2400	2.4
	Printer	40	2	8	640	0.64
	Copy machine	100	1	8	800	0.8
Subtotal						23.39
	The total daily energy demand of Kilu village					931.44

On the other hand, the NPC is calculated from equation (3)

$$\mathrm{NPC} = \frac{C_{ACC} + C_{ARC} + C_{AOM}}{\mathrm{CRF(i,\ n)}} \tag{9}$$

Where CRF (i, n) is the capital recovery factor, n is the number of years, and i is the real annual interest rate, which is calculated as follows;

$$\mathrm{i} = \frac{\mathrm{i}' - \mathrm{f}}{\mathrm{i}' + \mathrm{f}} \tag{10}$$

Where i' is the nominal interest rate, and f is the annual inflation rate (Das *et al* 2021, Kosov 2021).

3. Results and discussion

3.1. Load profile

To design an optimal standalone hybrid power system for Kiltu village, energy consumption was analyzed for households, the church, offices, and schools. Table 1 provides sector-wise usage, forming the basis for total load estimation. The total energy load for each service type was calculated using the equation:

$$L_S(\mathrm{kWh}) = \frac{1}{1000}\left(\sum_{n=1}^{N} Q_n P_n H_n\right) \tag{11}$$

Where L_S is the total load of service type in kilo watt, Q_n is the number of appliances in service type, P_n is the rated power of an appliance, H_n is the number of hours of operations.

This formula, adapted from (Amole *et al* 2023), calculates electrical demand by multiplying the number of appliances, their power ratings, and hours of usage. Using this method, the total daily energy consumption for Kiltu village was estimated to be 931.44 kWh, with a peak load of 115.19 kW, as shown in table 1. These estimates were essential for sizing the key components of the hybrid system, such as solar panels, wind turbines, batteries, diesel generators, and converters. The energy demand profiles used in the analysis were based on average consumption data from 400 households, a school, a church, and an administrative office, assuming typical and stable energy consumption patterns for rural areas. Solar irradiance and wind speed data from NASA, along with steady diesel supply data, were also incorporated into the study.

The RF was set to ensure sufficient system reliability, reducing reliance on diesel while meeting energy demand. Cost assumptions such as NPC and LCOE were based on current market rates for equipment, fuel, and maintenance, with expectations of stable costs throughout the system's lifespan. Environmental impacts, particularly CO_2 emissions, were calculated using standard emission factors for both diesel and renewable sources. It was also assumed that the system components solar panels, wind turbines, and batteries would follow typical industry lifespans and maintenance schedules.

3.2. Solar irradiation and wind speed analysis

The analysis of solar and wind resources in the study area reveals significant potential for renewable energy generation. The availability and variability of solar irradiation and wind speed were assessed using data of the region from NASA.

3.2.1. Solar irradiation analysis

The village receives an average daily solar irradiation of 6.06 kWh m^{-2}, with seasonal variations ranging from 5.23 kWh m^{-2} in July (the rainy season) to 6.57 kWh m^{-2} in February (the dry season). These fluctuations are crucial for system design, particularly when sizing the PV panels. During the rainy season, when solar output is lower, alternative energy sources such as wind power or diesel generators are essential to ensure a continuous power supply. As shown in figure 5 and table 3, the consistently high solar irradiation for most of the year positions as the primary energy source for village's hybrid power system.

3.2.2. Wind speed analysis

The study area experiences an average wind speed of 4.61 m s^{-1} at 50 meters, making it suitable for small to medium-scale wind turbines. Wind speeds fluctuate throughout the year, ranging from 3.11 m s^{-1} in September to 5.37 m s^{-1} in November. These variations highlight the importance of designing a hybrid energy system for the area. While the average wind speed supports energy generation, the dip in September emphasizes the need to integrate wind with other energy sources. As shown in figure 6 and table 3, wind energy reliably complements solar power, particularly during periods of low solar output

3.2.3. Combined solar and wind potential

The combination of solar irradiation and wind speed provides a strong foundation for a hybrid energy system at the study site. During months with lower solar radiation, such as July, moderate wind speeds help maintain a stable energy supply. In sunnier months, like February, solar power generation is enhanced. This seamless integration ensures a reliable electricity supply year-round, effectively addressing seasonal fluctuations in both resources. The analysis confirms the practicality of a solar-wind-diesel hybrid system, optimally designed to effectively meet the village's energy needs.

Table 2 presents hourly energy consumption data (0–23 h) for all 12 months, highlighting seasonal fluctuations in demand. Energy usage remains generally stable, with notable declines in July and August, likely due to reduced activity during the summer. Significant peaks occur between rows 6 and 19, indicating periods of high consumption. This analysis is crucial for optimizing energy supply during low-demand months, forecasting future demand, and addressing irregularities to enable more efficient resource management in the village.

The electricity load profile in the study area over a 24 h period is shown in figure 2. Key observations reveal distinct patterns in energy consumption throughout the day. From 1:00 to 6:00, energy use is minimal, reflecting low activity during the night.

A sharp rise in demand begins at 6:00, peaking near 50 kW by 9:00, driven by morning routines. The highest demand occurs between 11:00 and 13:00, reaching 110 kW, likely due to more intensive community activities. Energy use then dips to around 50 kW during the afternoon (14:00–17:00), indicating a quieter period. A second peak in demand occurs between 18:00 and 20:00, reaching 100 kW, corresponding to increased residential energy needs. After 20:00, energy consumption sharply declines, marking a return to low activity at night. This clear pattern, with two main demands peaks (midday and evening), reflects the village's daily rhythm and is valuable for energy supply planning, demand management, infrastructure design, and the efficient integration of renewable energy sources like solar power.

Figure 3 displays the monthly diurnal energy profiles (in kW) across a 24 h period, with individual subplots representing each month. The energy trends exhibit a consistent bimodal distribution, characterized by two prominent peaks in the morning and evening, likely driven by heightened energy demand from activities like cooking and lighting. Seasonal variations are evident, especially in the magnitude and timing of these peaks. For instance, July and August show lower peak values compared to other months, likely due to reduced energy demand influenced by factors such as climate, school closures, and changes in daily routines. From a systems

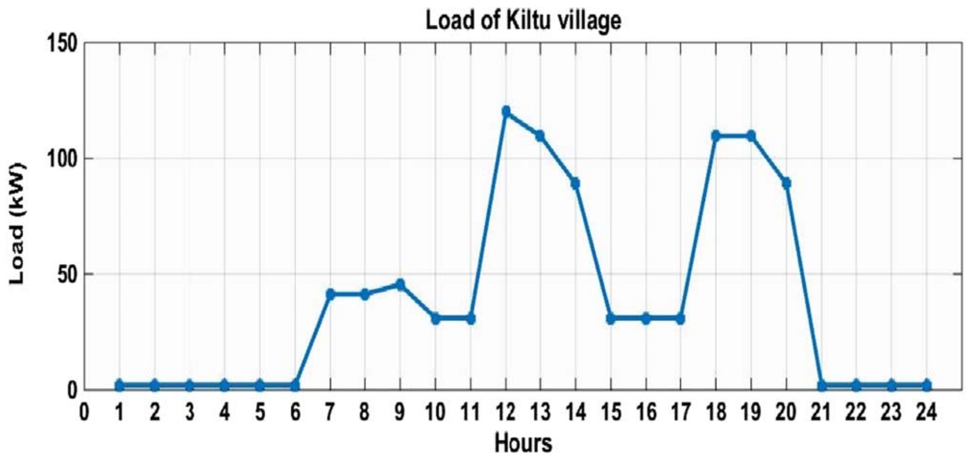

Figure 2. Load of the village.

Table 2. Hourly energy load of Kiltu Village (kWh).

Time	Jan.	Feb.	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
0	2.07	2.07	2.07	2.07	2.07	2.07	1.04	1.04	2.07	2.07	2.07	2.07
1	2.07	2.07	2.07	2.07	2.07	2.07	1.04	1.04	2.07	2.07	2.07	2.07
2	2.07	2.07	2.07	2.07	2.07	2.07	1.04	1.04	2.07	2.07	2.07	2.07
3	2.07	2.07	2.07	2.07	2.07	2.07	1.04	1.04	2.07	2.07	2.07	2.07
4	2.07	2.07	2.07	2.07	2.07	2.07	1.04	1.04	2.07	2.07	2.07	2.07
5	2.07	2.07	2.07	2.07	2.07	2.07	1.04	1.04	2.07	2.07	2.07	2.07
6	41.40	41.40	41.40	41.40	41.40	41.40	1.04	1.04	41.40	41.40	41.40	41.40
7	41.40	41.40	41.40	41.40	41.40	41.40	1.04	1.04	41.40	41.40	41.40	41.40
8	45.54	45.54	45.54	45.54	45.54	45.54	20.69	20.69	45.54	45.54	45.54	45.54
9	31.05	31.05	31.05	31.05	31.05	31.05	5.17	5.17	31.05	31.05	31.05	31.05
10	31.05	31.05	31.05	31.05	31.05	31.05	5.17	5.17	31.05	31.05	31.05	31.05
11	120.05	120.05	120.05	120.05	120.05	120.05	77.00	77.00	120.05	120.05	120.05	120.05
12	109.70	109.70	109.70	109.70	109.70	109.70	63.75	63.75	109.70	109.70	109.70	109.70
13	89.00	89.00	89.00	89.00	89.00	89.00	63.55	63.55	89.00	89.00	89.00	89.00
14	31.05	31.05	31.05	31.05	31.05	31.05	5.17	5.17	31.05	31.05	31.05	31.05
15	31.05	31.05	31.05	31.05	31.05	31.05	5.17	5.17	31.05	31.05	31.05	31.05
16	31.05	31.05	31.05	31.05	31.05	31.05	5.17	5.17	31.05	31.05	31.05	31.05
17	109.70	109.70	109.70	109.70	109.70	109.70	51.12	51.12	109.70	109.70	109.70	109.70
18	109.70	109.70	109.70	109.70	109.70	109.70	51.12	51.12	109.70	109.70	109.70	109.70
19	89.00	89.00	89.00	89.00	89.00	89.00	51.00	51.00	89.00	89.00	89.00	89.00
20	2.07	2.07	2.07	2.07	2.07	2.07	1.04	1.04	2.07	2.07	2.07	2.07
21	2.07	2.07	2.07	2.07	2.07	2.07	1.04	1.04	2.07	2.07	2.07	2.07
22	2.07	2.07	2.07	2.07	2.07	2.07	1.04	1.04	2.07	2.07	2.07	2.07
23	2.07	2.07	2.07	2.07	2.07	2.07	1.04	1.04	2.07	2.07	2.07	2.07

perspective, this visualization offers crucial insights into energy consumption and production trends over the year. It supports optimal resource allocation by identifying periods of peak demand, ensuring that energy storage and supply are effectively managed during these times.

For renewable energy systems, understanding these diurnal and seasonal fluctuations is critical for designing hybrid solutions that integrate complementary energy sources, enhancing both reliability and efficiency. These insights can also guide policy-making and strategic planning to strengthen energy resilience and sustainability in the study village and other rural areas.

The GHI peaks at 6.57 $kWh/m^2/day$ in February and drops to 5.23 $kWh/m^2/day$ in July, reflecting seasonal solar intensity variations. The clearness index is highest in December (0.69) and lowest in July (0.51), indicating clearer skies in winter and more atmospheric interference in summer. Wind speeds peak in November at 5.37 m s^{-1} and fall to 3.11 m s^{-1} in September, showing distinct seasonal patterns as detailed in table 3. These trends suggest strong solar potential from February to May and October to January, with reduced GHI in summer. Wind energy is most effective in winter, making a solar-wind hybrid system ideal for reliable, year-round energy production.

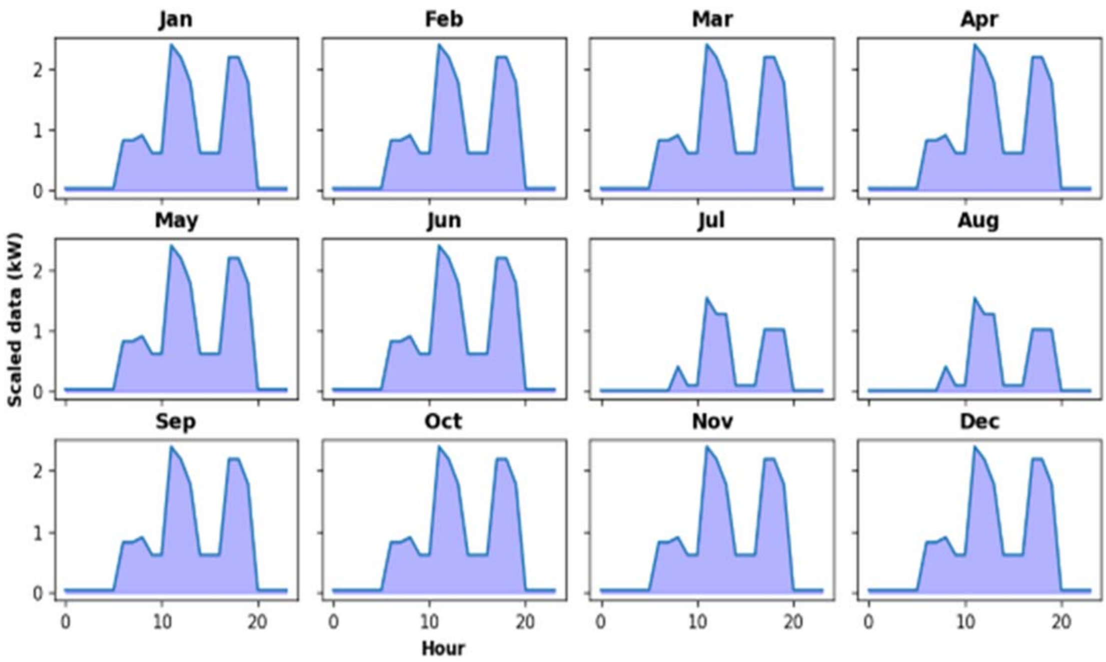

Figure 3. Scaled data daily profile.

Table 3. Average solar global horizontal irradiance GHI), clearness index, and wind speed.

Months	GHI (kWh/m^2/day)	Clearness index	Wind speed (m/s)
January	6.08	0.67	5.24
February	6.57	0.68	5.33
March	6.52	0.63	5.05
April	6.31	0.60	4.66
May	6.36	0.62	4.18
June	5.77	0.57	3.62
July	5.23	0.51	4.43
August	5.36	0.52	4.05
September	5.84	0.57	3.11
October	6.31	0.64	4.92
November	6.27	0.68	5.37
December	6.08	0.69	5.33

Figure 4 presents a dual-bar chart comparing GHI and the clearness index for each month. It ranges from 5.23 kWh/m^2/day to 6.57 kWh/m^2/day, while the clearness index is lower during the summer months (July to September) due to increased atmospheric interference, peaking in December, indicating clearer skies and more direct sunlight. Although solar potential remains strong year-round, the summer months exhibit slight reductions, emphasizing the need to optimize solar strategies for maximum efficiency throughout the year.

Figure 5 shows the monthly variation in wind speed at 50 meters. Wind speed peaks at 5.37 m s^{-1} in November, drops to 3.11 m s^{-1} in September, then rises to 5.33 m s^{-1} in February before declining again in June. Winds stabilize between 5.05 m s^{-1} and 5.55 m/s from December to March, and between 3.36 m s^{-1} and 4.66 m/s from April to August. This seasonal fluctuation indicates that wind power generation is most effective in winter (January, November, and December) and less effective in summer (June to September), emphasizing the need for hybrid energy systems.

This study outlines a systematic approach to designing renewable energy systems, emphasizing sustainability and cost-effectiveness through NPC and LCOE. The process includes data collection, load assessment, component selection, simulations, and technical-economic comparisons (figure 6). Component sizing is optimized for solar PV, wind turbines, battery storage, diesel generators, and converters based on local load

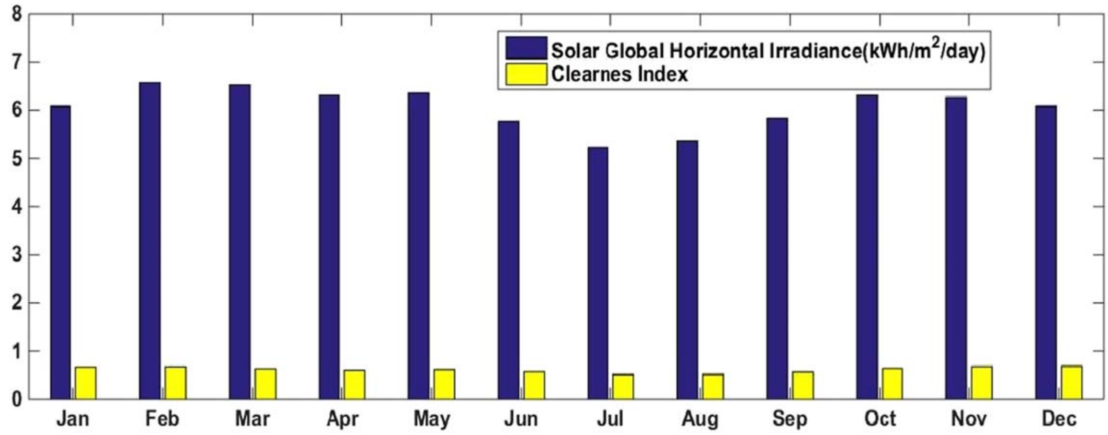

Figure 4. Solar global horizontal irradiance and clearness index.

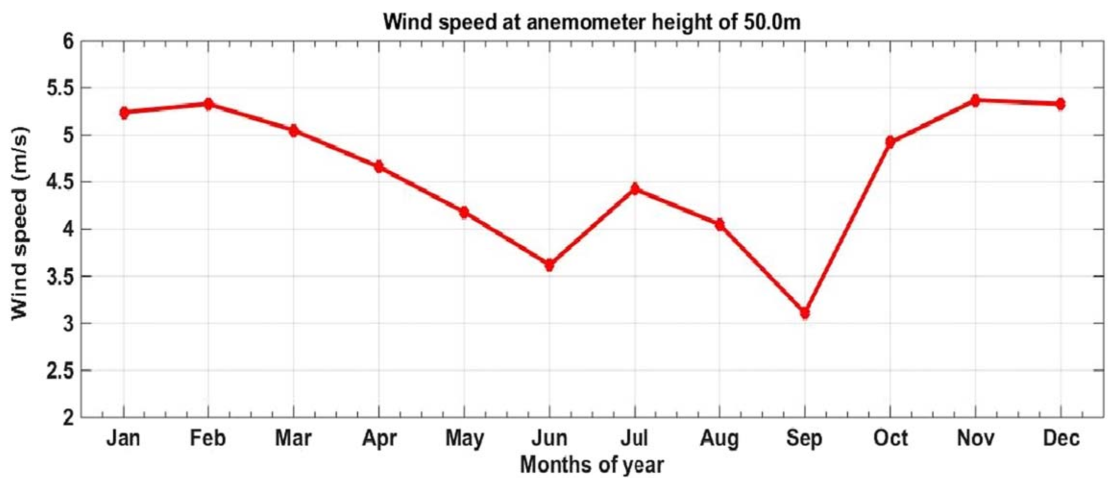

Figure 5. Wind speed at anemometer height of 50.0 m.

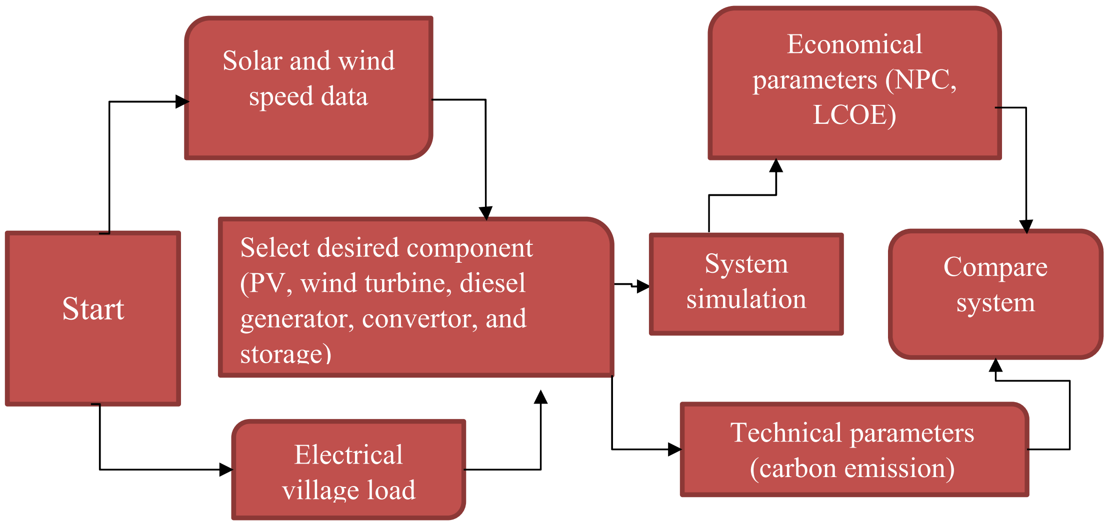

Figure 6. Presents a flowchart that outlines the system's analysis process.

Table 4. System configuration of hybrid energy system.

System	Setups	Reduction of CO_2 (%)
I	Gen//PV/battery/converter	49.16
II	Gen//PV//G3/battery/ converter	48.89
III	Gen//battery/converter	14.26
IV	Gen/G3/battery/converter	14.42
V	Gen	0
VI	Gen/PV	0.34
VII	G1/Gen	0.09
VIII	Gen/ PV/G3	0.77
IX	PV/ battery/converter	100
X	PV/G3/ battery/converter	100

profiles (931.44 kWh daily demands, 115.19 kW peak), resource availability, and capacity factors to ensure efficiency and reliability. The hybrid system integrates solar and wind to exploit complementary generation patterns, with battery storage for intermittency and a diesel generator as backup.

Solar PV meets daily demand and seasonal variations, while wind turbines are tailored to local conditions. Storage reduces fossil fuel reliance, and the converter minimizes losses during peak loads. This integrated system overcomes the limitations of standalone diesel or single renewable sources, enhancing reliability, reducing emissions, and stabilizing supply. It offers a sustainable, cost-effective solution for remote communities, providing resilient electrification without complex infrastructure.

Table 4 provides a comprehensive comparison of ten hybrid energy system configurations, evaluating their economic, operational, and environmental performance.

Economic Analysis: The systems display a wide range of costs, from the most economical system I (\$1.42 M) to the most expensive, system X (\$2.10 M). The higher costs of systems IX and X are attributed to their reliance on PV and G3, which require substantial upfront investment. Generator-based systems, like V (\$1.79 M) and VII (\$1.81 M), fall within the mid-cost range but offer minimal environmental benefits. In terms of energy production cost, system I leads with the lowest LCOE of \$0.326/kWh, while system X has the highest LCOE at \$0.499/kWh, reflecting the trade-off between zero emissions and higher financial requirements as shown in table 4.

Operational Cost: PV-integrated configurations, such as IX (\$59898) and X (\$60405), exhibit the lowest operational costs due to their reduced fuel use and low maintenance. In contrast, generator-based systems like V (\$134755) and VII (\$135266) incur higher operating expenses because of their fuel dependency and frequent maintenance needs.

Environmental Performance: System I emerges as the most cost-efficient choice, with the lowest LCOE and significant environmental benefits. Systems IX and X, while more expensive, offer zero emissions and are better suited for regions prioritizing sustainability. However, their higher financial costs could be a limiting factor. Generator-based systems, like V and VII, fail to deliver substantial CO_2 reductions, making them less viable for modern energy needs.

System I is a balanced option, achieving a 49.16% CO_2 reduction and System II provides similar benefits, with marginally lower emissions reduction (48.89%). These hybrid systems effectively balance both economic and environmental objectives, making them versatile for various energy demands. Both I and II are standout hybrids, combining low NPC, competitive LCOE, and significant emissions reductions, ideal for projects seeking both cost efficiency and environmental impact. Systems IX and X, while offering zero emissions, may be less feasible in regions with budget constraints due to their high NPC and LCOE. Generator-based systems (III, IV, V, VI, VII, and VIII) perform poorly across all metrics, with high emissions and operational costs, rendering them unsuitable for future energy needs.

Finally, System I is the best overall for cost, reliability, and moderate emissions reduction. System IX and X are ideal for carbon neutrality but requires a higher budget. Systems V–VIII should be avoided due to high costs, emissions, and inefficiency.

Figure 7 illustrates four energy system configurations for Kiltu village, labeled (a), (b), (c), and (d). These systems are designed to meet the village's energy demand of 931.44 kWh/day with a 115.19 kW peak load.

(a) Solar-generator hybrid system: The generator is the main power source, with solar PV providing additional energy. Excess solar power is stored in the battery, and a converter manages AC-DC conversion, reducing diesel use. Challenges include heavy diesel reliance, higher costs, and limited battery storage.

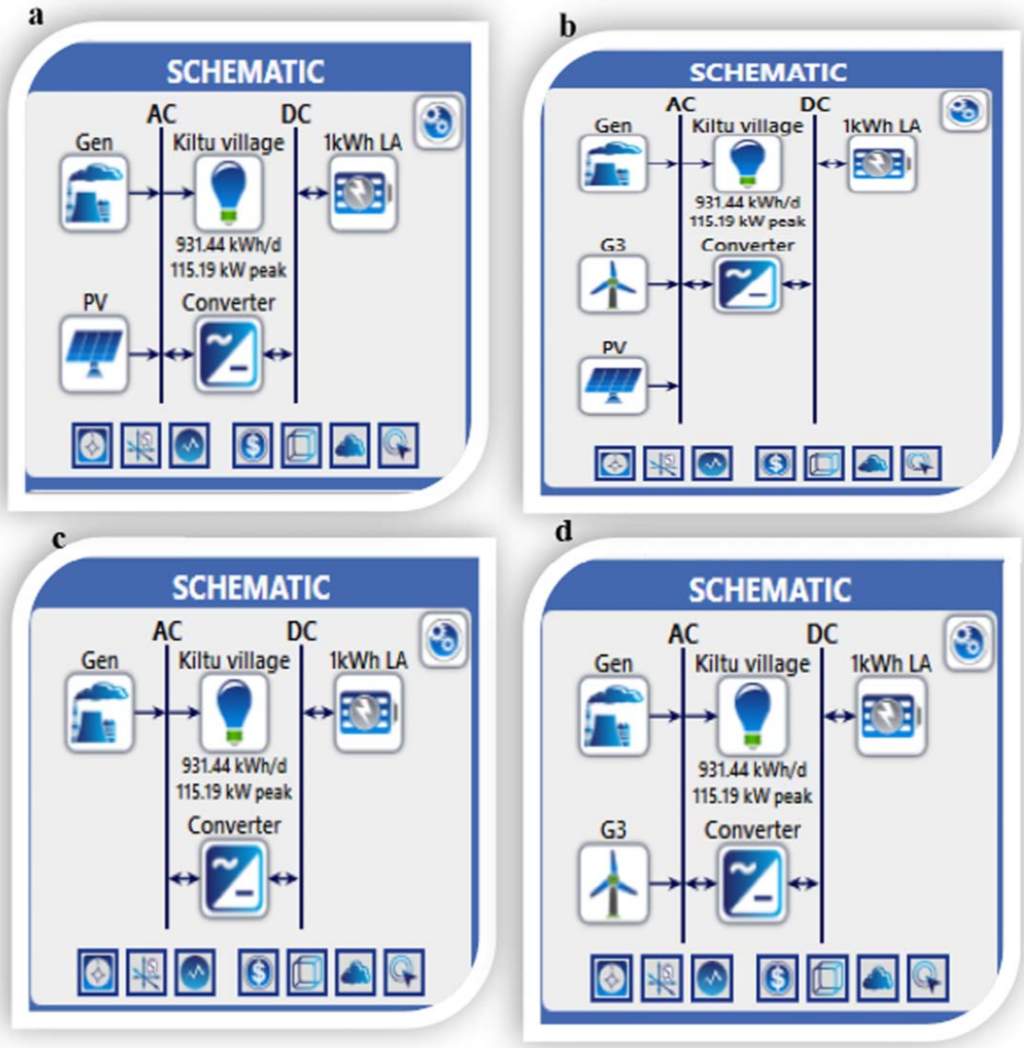

Figure 7. Design of the hybrid power system (a) Generator//PV/Battery/Converter and (b) Generator/G3/PV/Battery/Converter (c) Gen//battery/converter (d) Gen/G3/battery/converter configurations.

(b) Solar-wind-generator hybrid system: This system reduces diesel use by adding wind energy, especially when solar is low, such as at night. The generator backs up during peak loads. Benefits include improved reliability, lower costs, and reduced emissions. Challenges include the need for advanced controls and reliance on local wind conditions.

(c) Diesel generator-dominated system: This system consists of a diesel generator, a converter, and battery. The generator is the primary energy source, with the battery providing minimal backup for fluctuations. It's reliable but heavily reliant on diesel, leading to high costs and emissions, with no sustainability benefits.

(d) Wind- diesel generator hybrid system: This system integrates a diesel generator, wind turbine, converter, and battery as shown in figure 7. The wind turbine reduces diesel use, while the generator provides backup when wind is low. It offers fuel savings and improved sustainability but faces challenges from variable wind resources and limited battery storage.

Table 5 evaluates ten hybrid energy configurations by comparing their financial, operational, and environmental performance.

The PV/Gen/battery/converter design emerges as the most cost-effective, featuring the lowest NPC (\$1.42 M) and LCOE (\$0.326/kWh), along with moderate annual operating costs of \$77,738. In contrast, the PV/battery/converter and PV/G3/battery/converter configurations incur higher costs, with NPCs of \$2.10 M and LCOEs of \$0.499/kWh respectively, due to their dependence on PV and battery storage. Despite these

Table 5. Cost evaluation of hybrid energy system components.

Component	Capital($)	Replacement($)	O and M ($)	Fuel($)	Savage ($)	Total (&)
Autosize genset	65,000	159,472.70	167,961.79	419,943.24	5830.79	806,546.94
Generic 1kWh lead acid	70,500	217,325.30	30,040.89	0	12595.18	305271.01
Generic flat plate PV	268,764.53	0	11,452.38	0	0	280 216.90
System converter	17,619.85	7,352.15	0	0	1368.47	23,603.53
system	412,884.38	384,150.15	209,455.06	419943.24	19794.43	1415,638.39

higher initial costs, PV/battery/converter ($59,898/year) and PV/G3/battery/converter ($60,405/year) exhibit the lowest operating costs, benefiting from minimal fuel and maintenance requirements. Generator-based designs incur significantly higher operating costs ($134755/year) due to fuel consumption and frequent maintenance. Environmentally, the PV/battery/converter and G3/PV/battery/converter systems perform the best, with zero CO_2 emissions, making them ideal for sustainability-focused projects, despite their higher initial costs. In contrast, the generator-only scheme generates the highest CO_2 emissions (338258 kg/year), making it the least environmentally friendly option. The PV/Gen/Battery/Converter system achieves a 49.16% reduction in CO_2 emissions**,** closely aligning with the findings of (Samatar *et al* 2024), while maintaining a balance between cost-effectiveness, reliability, and environmental benefits. This conclusion is further reinforced by (Eze *et al* 2024).

The lifetime costs of a hybrid energy system, covering capital, replacement, O&M, fuel, and salvage values are outlined in table 5.

The Autosize genset is the most expensive component, totaling $806,546.94, primarily due to high fuel ($419,943.24) and replacement costs ($159472.70). Its negative salvage value (–$5,830.79) further adds to financial strain. The Lead Acid Battery has a high initial cost of $70,500 and the largest replacement expense at $217,325.30, bringing its total to $305,271.01. Despite having no fuel costs, frequent replacements and a negative salvage value (–$12,595.18) increase overall expenses. The PV System requires the highest upfront investment ($11,452.38) but has a lower lifetime cost of $280,216.90, highlighting its long-term cost-effectiveness. The System Converter is the most affordable at $23,603.53, with modest capital ($17,619.85) and replacement ($7,352.15) expenses, though its negative salvage value (–$1,368.47) slightly offsets savings.

The total system cost stands at $1,415,638.39, largely driven by fuel ($419,943.24) and replacement costs ($384,150.15) from the genset and battery. While diesel ensures reliability, it significantly raises operating costs, whereas solar energy minimizes ongoing expenses but demands higher initial investment. The negative salvage value (–$19,794.43) reflects long-term financial challenges. Achieving cost efficiency requires balancing renewable energy investments with conventional system expenses.

Kiltu Village's annual energy demand is 333,406 kWh, based on a daily consumption of 913.44 kWh. The proposed hybrid energy systems comfortably exceed this requirement, generating between 352,091 and 512,083 kWh annually. As shown in figure 8, the first system produces 379,072 kWh per year, resulting in a surplus of 19,353 kWh. Most configurations achieve zero unmet loads while also producing excess energy ranging from 2,010 to 123,232 kWh per year, ensuring ample storage capacity. Even the smallest surplus provides a reliable energy buffer. However, two systems experience minor unmet loads of 11,343–11,582 kWh annually, likely due to temporary inefficiencies in storage or distribution rather than insufficient generation. Overall, the hybrid systems not only fulfil the village's energy needs but also generate substantial excess energy up to 123,232 kWh supporting reliable storage and continuous electrification.

Hybrid systems combining solar PV, wind, diesel generators, and battery storage provide a solid mix of economic and environmental benefits. With a renewable energy share of 35.6% to 35.8%, a LCOE ranging from $0.326 to $0.332/kWh, and a NPC between $1.42 million and $1.44 million, they represent the most cost-effective solution. While the initial investment ($421,884 to $437,576) is higher than that of diesel-only systems, hybrid systems offer significantly lower annual operating costs ($77,000 to $78,000) and reduce CO_2 emissions (171,981 to 172,862 kg yr^{-1}), as indicated in figure 9. These advantages make them not only economically feasible but also environmentally sustainable, making them a perfect choice for off-grid communities in need of green energy, aligning with the findings of (Kumar *et al* 2025).

In contrast, diesel-only systems require lower initial investments ($65,000 to $139,721) but pose long-term challenges, such as higher LCOE values ($0.372 to $0.417/kWh) and NPCs ($1.62 million to $1.81 million). Their higher operating costs ($116,922 to $135,266/yr) and CO_2 emissions (290,023 to 338,258 kg/yr) make them unsustainable in the long run. Fully renewable systems, while emissions-free, demand high initial investments ($1.30 M to $1.32 M) and have a high NPC ($2.07 M to $2.10 M), despite lower operating costs ($59,898 to $60,405/yr). The feasibility of these systems depends on factors such as component costs ($300/kW

Architecture					System			Gen	PV	G3
					Elec Prod (kWh/yr)	Excess Elec (kWh/yr)	Unmet load (kWh/yr)	Production (kWh)	Production (kWh/yr)	Production (kWh/yr)
✓		✓	✓	✓	379,072	19,353	0	218,371	160,701	
✓	✓	✓	✓	✓	379,999	20,751	0	218,781	160,534	684
		✓	✓	✓	352,091	2,010	0	352,091		
	✓	✓	✓	✓	352,314	2,143	0	351,631		684
		✓			391,463	51,487	0	391,463		
✓		✓			391,770	51,794	0	389,593	2,177	
	✓	✓			391,619	51,643	0	390,935		684
✓	✓	✓			392,279	52,304	0	387,242	4,354	684
✓			✓	✓	510,889	122,256	11,582		510,889	
✓	✓		✓	✓	512,083	123,232	11,343		511,399	684

Figure 8. Comparative performance analyses of energy system architectures.

Architecture					Cost				System	
					NPC ($)	COE ($)	Operating cost ($/yr)	Initial capital ($)	Ren Frac (%)	CO_2 (kg/yr)
✓		✓	✓	✓	$1.42M	$0.326	$77,738	$421,884	35.8	171,981
✓	✓	✓	✓	✓	$1.44M	$0.332	$78,498	$437,576	35.6	172,862
		✓	✓	✓	$1.62M	$0.372	$116,922	$120,628	0	290,023
	✓	✓	✓	✓	$1.64M	$0.377	$117,258	$139,721	0	289,474
		✓			$1.79M	$0.411	$134,775	$65,000	0	338,258
✓		✓			$1.79M	$0.412	$134,567	$68,641	0	337,103
	✓	✓			$1.81M	$0.417	$135,266	$83,000	0	337,932
✓	✓	✓			$1.81M	$0.417	$134,855	$90,281	0	335,650
✓			✓	✓	$2.07M	$0.493	$59,898	$1.30M	100	0
✓	✓		✓	✓	$2.10M	$0.499	$60,405	$1.32M	100	0

Figure 9. Comparative analysis of renewable energy integration: economic costs and CO_2 emissions.

for solar, \$0.05 to \$0.10/kWh for batteries), diesel prices (\$1.00 to \$1.50/L), and discount rates (6% to 10%). A reduction in financing costs can greatly enhance the feasibility of both hybrid and renewable systems.

4. Conclusion

This study investigates the potential of a hybrid energy system combining solar, wind, and diesel backup for Kiltu Village, a remote, off-grid community in Oromia, Ethiopia. The region has significant renewable resources, with an average solar irradiation of 6.06 kWh/m^2/day, peaking at 6.57 kWh m^{-2} in February and dropping to 5.23 kWh m^{-2} during the rainy season in July, along with average wind speeds of 4.61 m s^{-1}, which peak in November. These seasonal variations underscore the importance of integrating solar panels, wind turbines, battery storage, and diesel generators to ensure a reliable energy supply year-round. By analyzing local energy demands, seasonal changes, and system configurations, the research shows that hybrid systems effectively balance cost efficiency and reliability. Solar-wind-battery solutions not only lower costs but also reduce reliance on biomass fuels, alleviating indoor air pollution and enhancing public health.

Simulations using HOMER Pro software identified an optimal hybrid configuration (solar PV, batteries, converters, and minimal diesel backup), achieving a LCOE of $0.326/kWh, a NPC of $1.42 million, and a 49.16% reduction in carbon emissions compared to diesel systems. These hybrid configurations, combining renewable sources with minimal diesel support, proved to be more sustainable and cost-effective than generator-dominated alternatives. To promote wider adoption of such solutions, the study recommends policy actions, such as subsidies for renewable energy technologies, improved energy monitoring systems, and further research to reduce technology costs for low-income communities. Expanding these approaches could accelerate Ethiopia's clean energy transition, reduce health risks in rural areas, and offer a model for off-grid communities globally seeking affordable and sustainable energy solutions.

CRediT authorship contribution statement

Tegenu Argaw Woldegiyorgis contributed to the conceptualization, methodology, software, formal analysis, writing of the original draft and the review and editing of the manuscript. **Eninges Asmare** was responsible for data curation, validation, investigation, and writing review and editing. **Natei Ermias Benti** contributed to visualization and writing review and editing. **Solomon Kebede Asefa** provided supervision and writing review and editing. **Abera Debebe Assamnew** was involved in data curation, validation, formal analysis, and writing review and editing. **Gezahegn Assefa Desalegn** contributed to software, formal analysis, and writing review and editing. **Fekadu Chekol** and **Fikru Abiko Anose** contributed to investigation and writing—review and editing. **Sentayehu Yigzaw Mossie** provided supervision and writing review and editing

Conflict of interest

The authors have no conflicts of interest to announce.

Declaration of competing interest

The authors declare that they have no competing financial interests that could have performed to influence this work.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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